

1. 解: 设观测器为 $\dot{Z} = (A+GC)Z + Bu - Gy$

$$C = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad A^T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad W = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W = (0 \ 1) R^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{G}^T = WR^T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow (0 \ 1)$$

$$\bar{A}^T = A^T + C^T \hat{G}^T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$|sI - \bar{A}^T| = s^2 - 1$$

$$(s+a)(s+2a) = s^2 + 3as + 2a^2$$

$$g^T = \begin{pmatrix} -1-2a^2 & -3a \end{pmatrix}$$

$$\tilde{g}^T = \begin{pmatrix} \bar{A}^T & C^T \end{pmatrix} \text{ 的能控性矩阵为: } \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

$$q = (0 \ 1) \quad T^{-1} = \begin{pmatrix} q \\ q\bar{A}^T \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\tilde{g}^T = g^T T^{-1} = \begin{pmatrix} -3a \\ -1-2a^2 \end{pmatrix} \begin{pmatrix} -3a & -1-2a^2 \end{pmatrix}$$

$$G^T = \hat{G}^T + \tilde{g}^T = \begin{pmatrix} -3a & -2a^2 \end{pmatrix}$$

$$G = \begin{pmatrix} -3a \\ -2a^2 \end{pmatrix}$$

则观测器为:

$$\dot{Z} = \begin{pmatrix} -3a & 1 \\ -2a^2 & 0 \end{pmatrix} Z + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u - \begin{pmatrix} -3a \\ -2a^2 \end{pmatrix} y$$

$$\text{判断能观性 } |sI - A| = 0 \quad \lambda = 0$$

$$\begin{pmatrix} \lambda I - A \\ C \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\text{rank} \begin{pmatrix} \lambda I - A \\ C \end{pmatrix} = 2 = n \quad \text{能观}$$

2. 解: ~~将改导原系统形式:~~

原系统形式:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

将其改写为:

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad A_{21} = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad A_{22} = 0$$

$$B_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad B_2 = 0$$

降维观测器: $\dot{w} = (A_{11} + GA_{21})w + B_1 u - (A_{11} + GA_{21})Gy$

求 G : 问题等效于设计 $\begin{cases} \dot{a} = A_{11}a + B_1 u \\ b = A_{21}a \end{cases}$ 的全维观测器中的 G

$$A_{21}^T = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad r=1 \quad n=2 \quad W = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad \hat{A}_{11}^T = A_{11}^T + GA_{21}^T \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = R^T$$

$$\hat{G}^T = WR^T = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad \bar{A}_{11}^T = A_{11}^T + GA_{21}^T \quad \hat{G}^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$(sI - \bar{A}_{11}^T) = s^2 - 1$$

$$(s+4)(s+5) = s^2 + 9s + 20$$

$$g^T = \begin{bmatrix} -21 & -9 \end{bmatrix}$$

$$g = \begin{bmatrix} 0 & 18 \end{bmatrix}$$

$$T^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\tilde{g}^T = g^T T^T = \begin{bmatrix} -9 & -21 \end{bmatrix} \quad \tilde{g} = \begin{bmatrix} -9 \\ -20 \end{bmatrix} \quad G = \begin{bmatrix} -9 \\ -20 \end{bmatrix}$$

则原系统降维观测器为:

$$\dot{w} = \begin{pmatrix} -9 & 1 \\ -20 & 0 \end{pmatrix} w + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u - \begin{pmatrix} 61 \\ 180 \end{pmatrix} y$$